

The Rise of On-Device Generative AI

Transforming Technology and User Experience



At present, on-device AI is revolutionising industries with real-time processing and enhanced user experiences. Processing data locally reduces latency, increases security, and allows for greater personalisation. This technology delivers immediate responses for applications like augmented (AR) and virtual reality (VR) while minimising data breach risks by keeping sensitive information on the device.

Generative AI: a new frontier

Generative AI, a subset of artificial intelligence, creates new content like text, images, and music-based on patterns from existing data. On-device generative AI is particularly transformative.

-Text generation: It can be used in applications like predictive text and unique content creation. For example, smartphones can generate custom responses in messaging apps or create reminders.

-Image and video generation: Devices with on-device generative AI enhance photos and videos in real-time, applying filters and improving quality.

-Music and sound: On-device AI can generate custom playlists and even compose new music tailored to the user's taste.

App developers can leverage generative AI to create dynamic content, automate tasks, and enhance user experiences, generating code, images, audio, and text for streamlined development and personalised interactions. This integration opens new avenues for innovation.

Generative AI is redefining mobile experiences with cutting-edge apps on Android and iPhone. Examples include Prisma, which transforms photos into artwork; Adobe Fresco for AI-driven art; and FaceApp for facial transformations. Amper Music creates AI-generated melodies, while Wysa offers AI-powered mental health chatbots. These apps utilise generative AI in the image and audio generation, natural language processing (NLP), and predictive modelling, showcasing its potential in the mobile landscape.

Platforms from leading AI chipmakers

- NVIDIA: NVIDIA's Jetson platform provides powerful AI computing for edge devices, supporting applications in robotics, healthcare, and smart cities. The Jetson Nano, for example, brings AI to small, power-efficient devices, making it accessible for developers and hobbyists alike.

- AMD: AMD's latest AI platform, the MI300, is designed to deliver high-performance AI and machine learning capabilities, suitable for various applications from data centres to edge devices³.

- Intel: Intel's Movidius Myriad X is a vision processing unit (VPU) for on-device AI, enhancing computer vision and deep learning in drones, cameras, and other devices.

Cloud service providers transforming generative AI

Cloud service providers like Amazon Web Services (AWS), Microsoft Azure, and Google Cloud are pivotal in advancing generative AI:

- Amazon Web Services (AWS): It provides tools to build and scale generative AI applications. Amazon Bedrock offers access to foundation models, while Amazon SageMaker allows developers to build, train, and deploy models. AWS's infrastructure supports high-performance AI workloads, facilitating the deployment of generative AI solutions.

- Microsoft Azure: Azure AI offers tools for building generative AI applications. Azure OpenAI Service provides access to powerful language models like GPT-4, enabling advanced conversational agents and content generation. Azure's infrastructure ensures scalability and reliability for AI applications.

- Google Cloud: Google Cloud's AI and machine learning services, including Vertex AI, offer tools for building and scaling AI models. It provides access to large language models like PaLM for generative AI applications. Google Cloud's infrastructure supports high-performance AI workloads, making it a key player in the generative AI landscape.

Leading large language models (LLMs)

Prominent large language models (LLMs) are reshaping artificial intelligence. Models like Claude (Anthropic), Haku (Google), and ChatGPT (OpenAI) excel in conversational understanding and generation. DALL-E (OpenAI) and Midjourney (Independent) create stunning images from text prompts, while LLaMA (Meta) offers efficient language modelling. BERT (Google) advances language understanding, and Stable Diffusion (Stability AI) provides flexible image generation. These LLMs fuel innovation in conversational AI, text generation, and language understanding. To address mobile constraints, smaller LLMs like MobileBERT, TinyBERT, and DistilBERT are emerging. Optimised for minimal memory and compute power, they use techniques like knowledge distillation,

pruning, and quantisation. These support AI applications such as virtual assistants, language translation, and text summarisation on resource-constrained devices.

AI integration in smartphones

AI integration in smartphones is rapidly advancing, with major manufacturers incorporating sophisticated AI features:

- Apple: The iPhone 16 series, powered by A18 chips, features on-device AI for real-time processing and personalisation, including writing tools and enhanced privacy.

- Google: The Pixel 8 Pro uses Gemini AI for Photo Unblur, Magic Editor, and Audio Eraser, enhancing editing capabilities.

- Samsung: The Galaxy S24 series includes Galaxy AI for advanced camera features and real-time translation.

- Other Manufacturers: Brands like Xiaomi, Oppo, Vivo, and Honor are also integrating generative AI into their flagship devices, from photography to voice assistance.

Porting large generative models to mobile devices

As demand for AI-powered mobile applications grew, developers faced challenges deploying generative models on non-GPU devices without compromising performance. Innovative architectures like MobileNet, ShuffleNet, SqueezeNet, and EfficientNet minimise size and computational needs. For instance, MobileNet uses depth-wise separable convolutions and linear bottlenecks for efficient computation. These architectures enable fast inference times and allow developers to seamlessly integrate generative models, unlocking immersive experiences like image and audio generation on resource-constrained devices.

The future of on-device generative AI

On-device generative AI is set to transform industries by enabling smarter, faster, and more secure solutions. With leaders like Apple, Google, and Samsung, alongside cloud support from AWS and Azure, this technology will redefine our daily lives and capabilities in smartphones, cars, and wearables.

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